

SmartCare: One Event, Three Responses

Student name: _____

Section: _____

System name: SmartCare

Selected action: typing a password on the Sign Up page.

Event: password:evaluated

The screenshot displays the SmartCare Sign Up page. On the left, a dark blue sidebar contains the SmartCare logo, a headline 'Simplifying healthcare, one appointment at a time.', a sub-headline 'SmartCare keeps patients and clinic staff in sync — clear schedules, live queues, no waiting-room guesswork.', a 'LIVE QUEUE MONITORING' section with a line graph, and a list of features: 'Easy appointment scheduling', 'Real-time queue management', and 'Convenient patient access'. The main content area is light gray and contains a 'Log In' button, a 'Sign Up' button, and a 'Create your SmartCare account' section. This section includes a sub-headline 'Set up secure access in under a minute.' and four input fields: 'Full name' (filled with 'Juan Dela Cruz'), 'Email address' (filled with 'you@example.com'), 'Phone number' (filled with '0917 123 4567'), and 'Password'. The password field is filled with dots. Below the password field, a strength indicator shows 'Very Strong' with a teal bar. To the right of the bar, a checklist of requirements is shown with teal circles indicating which are met: '8+ characters', 'One lowercase letter', 'One special character', 'One uppercase letter', and 'One number'. Below the strength indicator is a 'Confirm password' field filled with 'Re-enter password'. At the bottom of the sign-up section is a checkbox for 'I agree to the Terms and Privacy Policy.' and a large blue 'Create Account' button. Below the button is a link 'OR SIGN UP WITH' and a note about Google accounts. At the very bottom, there is a link 'Already have an account? Log in' and a 'SECURE PORTAL' link.

SmartCare

Log In Sign Up

Create your SmartCare account

Set up secure access in under a minute.

Full name
Juan Dela Cruz

Email address
you@example.com

Phone number
0917 123 4567

Password
.....

Very Strong

- 8+ characters
- One lowercase letter
- One special character
- One uppercase letter
- One number

Confirm password
Re-enter password

☐ I agree to the Terms and Privacy Policy.

Create Account

OR SIGN UP WITH

Google accounts are created as patient accounts. You can add your phone number later.

Already have an account? [Log in](#)

SECURE PORTAL

Figure 1. When the user types a password, SmartCare checks it and updates the checklist, strength bars, and strength label. This helps the user see which password rules they have passed before creating an account.

Three handlers respond to the same event

Each handler has a different job. All three listen to `password:evaluated`.

Handler	What it does	Data it uses
Handler 1	Marks the password rules that passed.	<code>checks</code>
Handler 2	Fills the strength bars.	<code>metCount</code> , <code>empty</code>
Handler 3	Changes the strength label and its color.	<code>metCount</code> , <code>empty</code>

js/password-strength.js — three anonymous handlers

```
15  events.subscribe("password:evaluated", ({ checks }) => {
16    Object.keys(checks).forEach((rule) => {
17      const item = strengthBlock.querySelector(`.req-list li[data-rule="${rule}"]`);
18      if (item) item.classList.toggle("met", checks[rule]);
19    });
20
21    }, "Update password requirements");
22
23  events.subscribe("password:evaluated", ({ metCount, empty }) => {
24    strengthBars.forEach((bar, index) => {
25      const isFilled = !empty && index < metCount;
26      bar.classList.toggle("filled", isFilled);
27      bar.style.setProperty("--dot-color", isFilled ? STRENGTH_COLORS[metCount - 1] : "");
28    });
29    }, "Update strength bars");
30
31  events.subscribe("password:evaluated", ({ metCount, empty }) => {
32    strengthLabel.textContent = empty ? "Password strength" : STRENGTH_NAMES[Math.max(metCount - 1, 0)];
33    strengthLabel.style.color = empty ? "" : STRENGTH_COLORS[Math.max(metCount - 1, 0)];
34    }, "Update strength label");
```

Figure 2. The three `events.subscribe` calls in `js/password-strength.js` register Handler 1, Handler 2, and Handler 3, from top to bottom.

Where is the anonymous or lambda function?

Each handler uses an arrow function with the `=>` symbol. For example, Handler 1 starts with `({ checks }) => {`. It has no function name. It runs when the event is published and updates the checklist.

The labels such as “Update password requirements” describe the handlers in the console. They are not function names. The screenshot shows the current source code displayed with line numbers.

What data is passed with the event?

Event data	Meaning
checks	The pass or fail result for each password rule.
metCount	The number of rules passed.
empty	Whether the password field is empty.

js/password-strength.js — action and event data

```

36   function evaluate(value) {
37       const checks = SmartCareValidators.checkPasswordRules(value);
38       const metCount = Object.values(checks).filter(Boolean).length;
39       events.publish("password:evaluated", { checks, metCount, empty: value.length
=== 0 });
40       return checks;
41   }
42

53   passwordInput.addEventListener("input", () => {
54       evaluate(passwordInput.value);
55       if (window.SmartCareFormHelpers) SmartCareFormHelpers.showError(passwordInput,
false);
56
57       if (confirmInput?.value && window.SmartCareFormHelpers) {
58         const mismatch = confirmInput.value !== passwordInput.value;

```

Figure 3. The input listener calls `evaluate`. This function checks the password and passes `{ checks, metCount, empty }` with the event. The raw password is not passed in this event.

js/event-center.js — invoking the registered handlers

```

60   function publish(eventName, data = {}) {
61       const handlers = subscribers[eventName] || [];
62       console.log(`[${eventName}] Event published - ${handlers.length} handlers`);
63       console.log(`[${eventName}] Event data: ${SmartCareEventCenter.describeData(data, eventName)}`);
64       handlers.forEach((entry) => {
65           try {
66               entry.handler(data);
67               console.log(`[${eventName}] Completed: ${entry.label}\n`,
`Event data: ${SmartCareEventCenter.describeData(data, eventName)}\n`);
69           } catch (error) {
70               console.error(`[${eventName}] Failed: ${entry.label}`, error);
71           }
72       });
73   }
74

```

Figure 4. The event center calls each registered handler using `entry.handler(data)`. After the handler returns, it prints a completion message. This shows that one event can call several functions.

The console shows the handlers running

I opened the browser Console, cleared the old messages, and entered a password in the Sign Up form. The screenshot below shows the result.

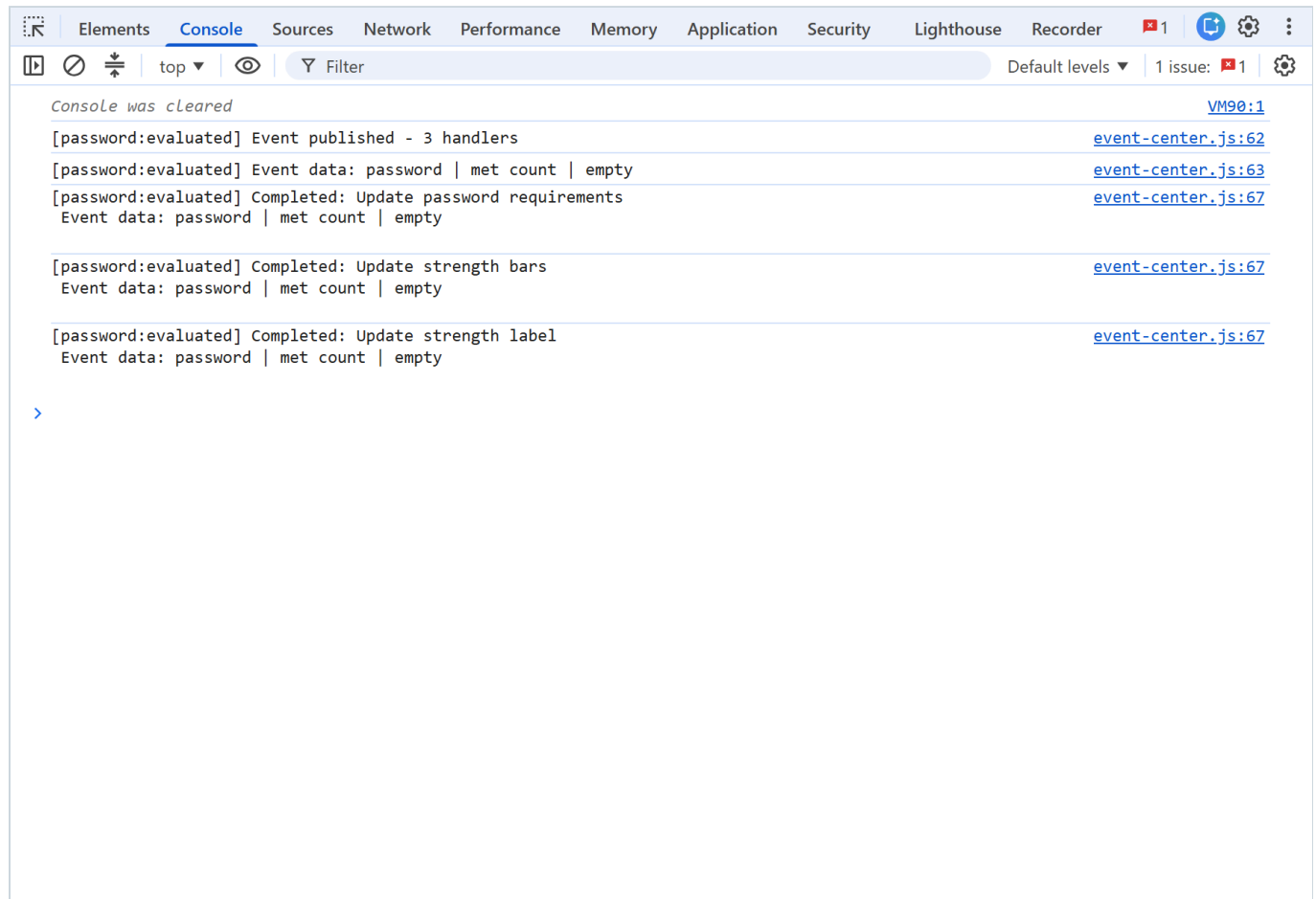


Figure 5. The actual browser Console shows one `password:evaluated` event and three completed handlers. “Update password requirements” is Handler 1, “Update strength bars” is Handler 2, and “Update strength label” is Handler 3.

The console uses short labels for the data. Here, `password` means the password checks, `met count` means the number of passed rules, and `empty` means whether the field is empty. The exact field names and their use are shown on pages 2 and 3.

Short reflection

This feature shows how one action can cause several useful responses. When a password is entered, one event passes the check results to three handlers, and each handler updates a different part of the page. Using separate handlers makes the code easier to follow and lets the same password indicator be used in other forms.

This demonstration uses the existing SmartCare system. No test account was created. The system and Console images are browser screenshots; the code images show the existing source displayed for reading.